

4. Experiments

This section evaluates the two-stage AE-SVDD defense under a fixed and reproducible federated-learning protocol. The design keeps three objects separate: the client-selection score, the AE-SVDD training objective, and the final aggregation weights. Numerical entries remain blank until they are produced by complete, parseable result files.

4.1 Experimental Setup

4.1.1 Datasets and Models

The primary study covers four image tasks and one text task. A fixed clean server-validation subset is created before client partitioning; its samples are withheld from all clients.

Dataset	Input and classes	Global model	Purpose
MNIST	28 x 28 grayscale, 10 classes	LeNetClassifier	Basic image-classification sanity check
Fashion-MNIST	28 x 28 grayscale, 10 classes	FashionCNN	Fine-grained grayscale classification and stability check
CIFAR-10	32 x 32 RGB, 10 classes	CIFAR-adapted ResNet-18	Deeper visual model and primary robustness task
COVID-19 Radiography Database	Chest X-ray images, 4 classes	Lightweight CNN	Medical-image classification and cross-domain robustness
AG News	Tokenized text, 4 classes	Lightweight Transformer classifier	Cross-modal transfer to text

The COVID-19 task follows the commonly used four-class COVID-19 Radiography Database protocol. Its loader, preprocessing, client partition, and model calibration must be recorded before any result is included in the primary matrix. Accuracy is compared only within the same dataset. Cross-dataset conclusions use relative accuracy drop, detection quality, and overhead rather than raw accuracy. Image-trigger ASR is `N/A` for AG News.

4.1.2 Federated Learning Protocol

The following settings are shared by every defense and attack unless a subsection explicitly varies one factor.

Setting	Primary value
Total clients, K	100
Malicious clients	30 (30%; identities are used only for post-hoc scoring)
Participation	All clients participate in every round
Communication rounds	300 for confirmation; 100 for screening
Local epochs / batch size	1 / 64
Client optimizer	SGD, momentum 0.9
Data partition	Dirichlet alpha = 1.0; IID is the clean reference condition
Trusted validation set	50 clean, stratified training samples withheld from clients

Task calibration is performed once with clean FedAvg using TACC only. The selected client optimizer settings are then shared by all defenses.

Dataset	<code>client_lr</code>	<code>client_weight_decay</code>
MNIST	0.10	1e-4
Fashion-MNIST	0.10	0
CIFAR-10	0.05	1e-4
COVID-19 Radiography Database	To be calibrated	To be calibrated
AG News	0.10	0

4.1.3 Attack Settings

The attack implementations are modular client components under `src/attacks`. Values below are fixed unless an intensity experiment changes the named parameter.

ID	Family	Upload or data transformation	Primary value
None	Control	Ordinary local training	No malicious clients
LF	Data poisoning	Symmetric label map, $y' = c - 1 - y$	Fixed by the task label space
GN	Byzantine poisoning	Moment-matched Gaussian replacement of floating tensors	<code>gaussian_sigma = 0.3</code>
SF	Byzantine poisoning	Upload $\bar{w}_g - s(w_l - \bar{w}_g)$	<code>sign_flip_scale = 1.0</code>
LIE	Byzantine poisoning	Craft delta as $\mu + z \text{ sigma}$ from benign updates	<code>lie_z_override = 0.524</code>
BD	Backdoor	Lower-right square trigger, target label, and model replacement	target 0; poison 0.6; trigger 5; value 1.0; replacement 3.0
Mix (M1)	Mixed attack	Deterministic round-robin assignment across malicious clients	<code>lf, bd, gn</code>

Mix is simultaneous: different malicious clients apply different attacks in the same round. It is not a random choice of one attack per round. The assignment map is saved with every result file so that overall and per-family recall can be computed. Supplementary combinations are M2=`lf,sf,lie` and M3=`lf,bd,gn,lie`; they are reported only when matched runs are available. The primary AG News matrix excludes image-trigger BD and Mix.

4.1.4 Defense Settings

All methods receive the same client states, data partitions, round budget, and information boundary. No baseline receives malicious identities or clean validation labels unless its protocol explicitly uses the shared validation set.

Method	Code ID	Core operation	Parameter settings
FedAvg	<code>avg</code>	Uniform aggregation of all uploads	No defense-specific parameter
Trimmed Mean	<code>tm</code>	Coordinate-wise trimmed aggregation	<code>trimmed_mean_ratio=0.2</code> ; <code>trimmed_mean_num_byzantine=null</code> (inferred from client composition)
Multi-Krum	<code>mk</code>	Distance-based Byzantine selection	<code>krum_num_byzantine=null</code> ; <code>multi_krum_num_selected=null</code> (use the implementation defaults)
LASA	<code>lasa</code>	Layer-adaptive sparsified aggregation	<code>lasa_sparsity_ratio=0.9</code> ; <code>lasa_lambda_n=1.0</code> ; <code>lasa_lambda_s=1.0</code>
FedSECA	<code>seca</code>	Sign election and coordinate aggregation	<code>fedseca_sparsity_gamma=0.9</code> ; <code>fedseca_temperature=1.0</code>
BNGuard	<code>bnguard</code>	Robust BN-feature distance filtering	<code>bnguard_tau=3.0</code>
FedDMC-style	<code>dmc</code>	Magnitude, direction, sign, sparsity, and temporal views	<code>dmc_warmup_rounds=3</code> ; <code>dmc_tau=3.0</code> ; <code>weights = (1.0, 1.0, 1.0, 0.5, 1.0)</code> for norm/direction/sign/sparsity/temporal
AE-SVDD (ours)	<code>svdd</code>	Fixed descriptor, AE reconstruction, and latent compactness	<code>latent_dim=64</code> ; <code>phase1_rounds=15</code> ; Phase 1= recon; Phase 2= combined; <code>alpha=0.5</code> ; descriptor dim= 4096
FL-Defender	<code>fld</code>	PCA/reputation-based update detector	<code>fldefender_pca_components=2</code> ; <code>fldefender_q1=0.25</code>
AlignIns	<code>alignins</code>	Direction and principal-sign alignment	<code>alignins_sparsity=0.9</code> ; <code>alignins_lambda_s=1.0</code> ; <code>alignins_lambda_c=1.0</code>
FLGMM / FLANDERS	<code>flgmm</code> / <code>flanders</code>	Registered comparison implementations	FLGMM: <code>warmup_rounds=50</code> , <code>control_l=3.0</code> , <code>em_iters=50</code> ; FLANDERS: <code>window=5</code> , <code>sampling=500</code> , <code>maxiter=100</code>

Supplementary defenses are included only when their JSON contains the same task, seed, round budget, client population, attack, partition, and validation protocol as the comparison slice.

4.1.5 Evaluation Metrics

Global utility

- **TACC**: clean test accuracy of the final global model. In tables, `acc` is the display label for TACC.
- **ASR**: fraction of triggered test images predicted as the target label. TACC and ASR are reported together for image backdoors and mixed image attacks.

Client detection

- **DAR**: the fraction of all participating clients that are correctly classified as benign or malicious.
- **DPR**: the fraction of rejected clients that are truly malicious, measuring the reliability of the rejection decision.
- **RR**: the fraction of malicious clients that are correctly rejected; this is equivalent to the malicious-client recall or true-positive rate (TPR).

5. Results and Analysis

This section reports the results using the same task, partition, attack, defense, and seed slices defined in Section 4. Numerical values are inserted only from complete, parseable result files. Word tables remain editable portrait tables.

5.1 Overall Performance

5.1.1 Global Model Utility

The primary matrix compares the eight primary defenses under the supported attacks. The portrait Word table groups conditions as Data Poisoning (LF), Byzantine Poisoning (GN, SF, LIE, Min-Max), Backdoor Attacks (BD, DBA), and Mix (M1, M2). `Min-Max`, `DBA`, and `M2` remain explicit columns and must be marked pending or `N/A` unless matched result files exist. `ACC` denotes TACC; `ASR` is reported for BD/DBA and the corresponding mixed conditions.

Table 1. Global model utility and attack suppression.

Dataset (Model)	Defense Method	No Attack	Data Poisoning	Byzantine Poisoning				Backdoor Attacks		Mix			
		ACC	LF	GN	SF	LIE	Min-Max	BD	DBA	M1		M2	
			ACC	ACC	ACC	ACC	ACC	ASR	ASR	ACC	ASR	ACC	ASR
MNIST (LeNet)	FedAvg	—	—	—	—	—	—	—	—	—	—	—	—
	Trimmed Mean	—	—	—	—	—	—	—	—	—	—	—	—
	Multi-Krum	—	—	—	—	—	—	—	—	—	—	—	—
	LASA	—	—	—	—	—	—	—	—	—	—	—	—
	FedSECA	—	—	—	—	—	—	—	—	—	—	—	—
	BNGuard	—	—	—	—	—	—	—	—	—	—	—	—
	FedDMC	—	—	—	—	—	—	—	—	—	—	—	—
	AE-SVDD (Ours)	—	—	—	—	—	—	—	—	—	—	—	—
Fashion-MNIST (CNN)	FedAvg	—	—	—	—	—	—	—	—	—	—	—	—
	Trimmed Mean	—	—	—	—	—	—	—	—	—	—	—	—
	Multi-Krum	—	—	—	—	—	—	—	—	—	—	—	—
	LASA	—	—	—	—	—	—	—	—	—	—	—	—
	FedSECA	—	—	—	—	—	—	—	—	—	—	—	—
	BNGuard	—	—	—	—	—	—	—	—	—	—	—	—
	FedDMC	—	—	—	—	—	—	—	—	—	—	—	—
	AE-SVDD (Ours)	—	—	—	—	—	—	—	—	—	—	—	—
CIFAR-10 (ResNet-18)	FedAvg	—	—	—	—	—	—	—	—	—	—	—	—
	Trimmed Mean	—	—	—	—	—	—	—	—	—	—	—	—
	Multi-Krum	—	—	—	—	—	—	—	—	—	—	—	—
	LASA	—	—	—	—	—	—	—	—	—	—	—	—
	FedSECA	—	—	—	—	—	—	—	—	—	—	—	—
	BNGuard	—	—	—	—	—	—	—	—	—	—	—	—
	FedDMC	—	—	—	—	—	—	—	—	—	—	—	—
	AE-SVDD (Ours)	—	—	—	—	—	—	—	—	—	—	—	—
AG News (Transformer)	FedAvg	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	Trimmed Mean	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	Multi-Krum	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	LASA	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	FedSECA	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	BNGuard	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	FedDMC	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	AE-SVDD (Ours)	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A

5.1.2 Malicious Client Detection

Detection is evaluated independently from global utility. The compact portrait Word table reports DAR / DPR / RR in every selected attack column. The complete analysis additionally retains FPR, F1, AUROC, AUPRC, accepted-client fraction, selected rejection ratio, candidate validation accuracies, losses, and mixed-attack per-family recall.

Table 2. Malicious client detection under different attacks.

Dataset (Model)	Detector	Data Poisoning			Byzantine Poisoning									Backdoor Attacks					
		LF			GN			SF			Min-Max			BD			DBA		
		DAR	DPR	RR	DAR	DPR	RR	DAR	DPR	RR	DAR	DPR	RR	DAR	DPR	RR	DAR	DPR	RR
MNIST (LeNet)	FedAvg	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	Trimmed Mean	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	Multi-Krum	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	LASA	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	FedSECA	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	BNGuard	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	FedDMC	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	AE-SVDD (Ours)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Fashion-MNIST (CNN)	FedAvg	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	Trimmed Mean	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	Multi-Krum	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	LASA	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	FedSECA	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	BNGuard	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	FedDMC	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	AE-SVDD (Ours)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CIFAR-10 (ResNet-18)	FedAvg	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	Trimmed Mean	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	Multi-Krum	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	LASA	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	FedSECA	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	BNGuard	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	FedDMC	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
	AE-SVDD (Ours)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
AG News (Transformer)	FedAvg	—	—	—	—	—	—	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	Trimmed Mean	—	—	—	—	—	—	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	Multi-Krum	—	—	—	—	—	—	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	LASA	—	—	—	—	—	—	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	FedSECA	—	—	—	—	—	—	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	BNGuard	—	—	—	—	—	—	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	FedDMC	—	—	—	—	—	—	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A
	AE-SVDD (Ours)	—	—	—	—	—	—	—	—	—	—	—	—	N/A	N/A	N/A	N/A	N/A	N/A

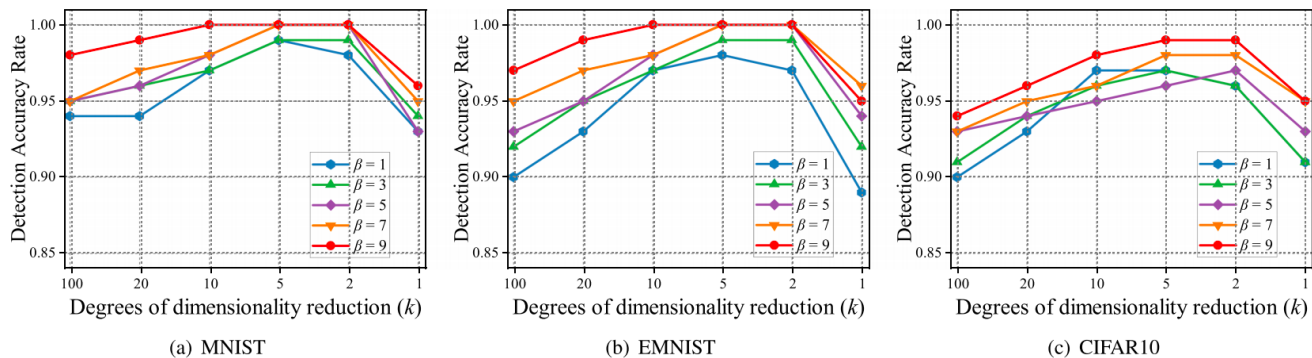
5.2 Parameter Sensitivity

Sensitivity experiments vary one AE-SVDD factor at a time. Unless a subsection changes it, the fixed configuration is descriptor dimension 4096, latent dimension 64, Phase-1 length 15, Phase-1 score `recon`, Phase-2 score `combined`, and alpha = 0.5. The screening budget is 100 rounds; selected settings are re-run for 300 rounds with seeds 42--44 before supporting primary claims.

5.2.1 Loss Coefficient

Sweep `alpha` in `{0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8}`. Values 0 and 1 may be included as diagnostic pure-reconstruction and pure-SVDD controls, but are not required for the main sensitivity claim. Keep selection scores fixed (`recon` then `combined`) so only the Phase-2 training objective changes.

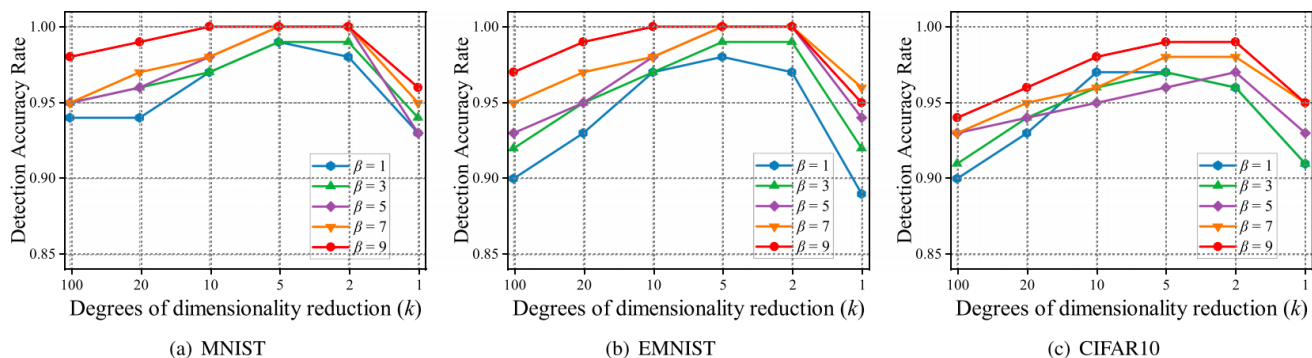
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5.2.2 Phase-1 Duration

Sweep `phase1_rounds` in `{5, 15, 30, 50}` with total rounds fixed at 100 for screening. Phase 1 always uses reconstruction-error ranking, and Phase 2 always uses the combined reconstruction-error plus SVDD-distance ranking. This isolates the duration of the reconstruction warm-up.

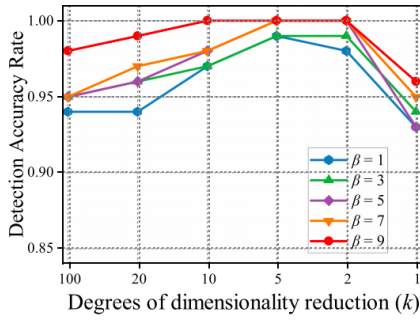
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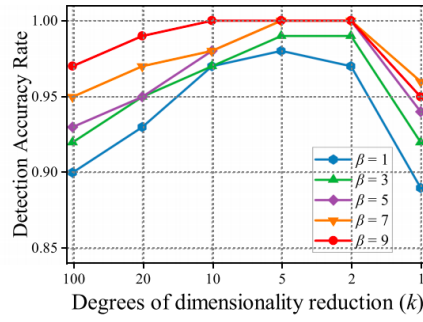
5.2.3 Trusted Validation-Set Size

Vary the direct sample count `server_validation_size` in `{10, 25, 50, 100, 200}`. Every validation set is clean, stratified, and withheld from clients. The primary value is 50 direct samples. Keep the candidate rejection grid and tie rule unchanged so only validation reliability changes.

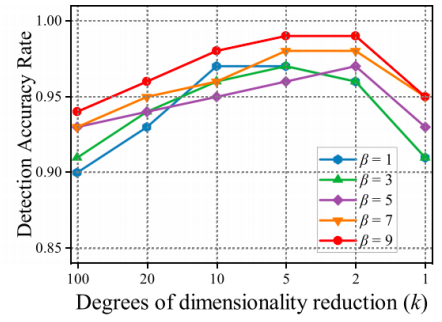
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(a) MNIST



(b) EMNIST

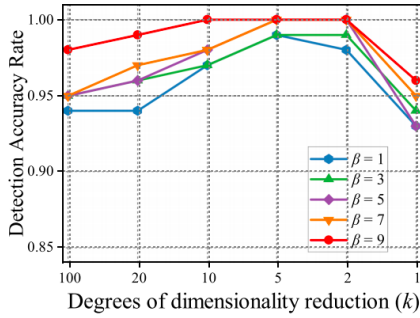


(c) CIFAR10

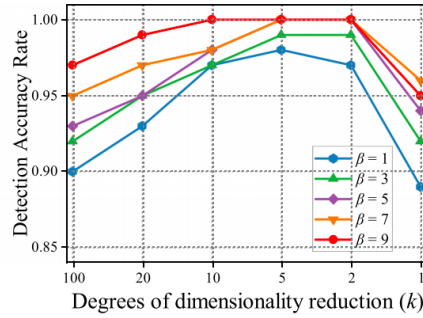
5.2.4 Latent Representation Dimension

Sweep `latent_dim` in `{16, 32, 64, 128}` while keeping the input descriptor dimension and all optimization settings fixed. This evaluates the compression bottleneck without changing the client-selection rule.

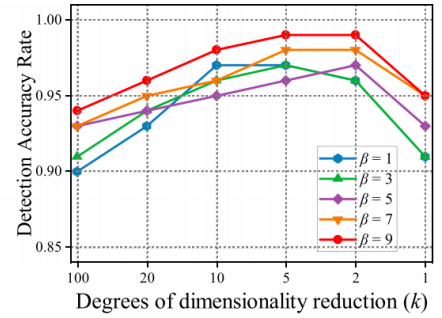
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(a) MNIST



(b) EMNIST



(c) CIFAR10

5.3 Robustness Analysis

Robustness factors are environment or attacker factors, not AE-SVDD training hyperparameters. Unless stated otherwise, use the 300-round primary schedule and report the final-10-round mean with three seeds.

5.3.1 Malicious-Client Ratio

Use malicious ratios of 10%, 20%, 30%, and 40%, with a 0% clean control. Keep `num_clients=100` and adjust `num_malicious` accordingly. A 50% boundary point may be shown separately, but it is outside the benign-majority claim.

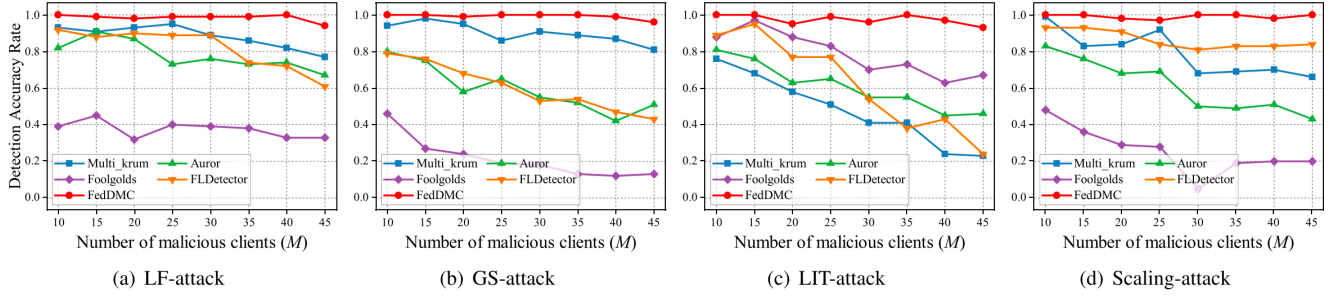


Fig. 5. Impact of the number of malicious clients on the defense methods under different attacks.

5.3.2 Data Heterogeneity

Compare IID (`dirichlet_alpha=null`) with Dirichlet alpha values 1.0, 0.5, and 0.1, using 100 clients and 30% malicious clients. Report benign FPR separately from malicious RR so normal client drift is not mistaken for attack detection.

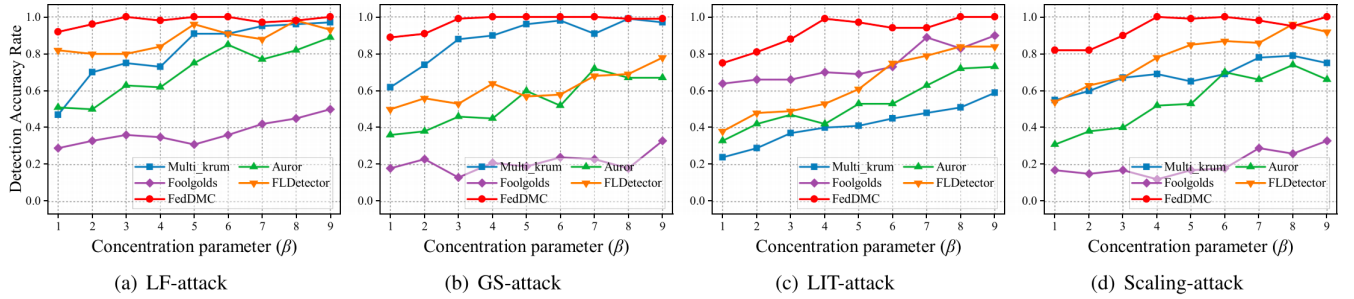


Fig. 6. Impact of the degree of non-IID on the EMNIST dataset under different attacks.

5.4 Ablation Study

All ablations keep the task, partition, seed, client count, attack parameters, validation size, local training budget, and candidate rejection grid fixed. Only the named component changes. Following standard component-ablation practice, `Full` is the reference configuration and each reduced configuration removes one stage. Comparisons are paired by the same dataset, malicious ratio, attack, and seed; cross-condition averages are not used to claim component complementarity.

5.4.1 Two-Stage Ranking Schedule

The experiment retains exactly three configurations. The names P1-only, P2-only, and Full are kept for continuity, while the parenthetical labels make the removed component explicit:

Configuration	Standard label	Phase-1 ranking	Phase-2 ranking	Interpretation
P1-only	without Phase 2	Reconstruction error in all rounds	Not entered	Reconstruction-only ranking
P2-only	without Phase 1 warm-up	Not entered	Combined reconstruction error + SVDD distance in all rounds	Combined-score ranking without warm-up
Full	complete model	Reconstruction error in the first 15 rounds	Combined reconstruction error + SVDD distance thereafter	Complete two-stage ranking

P1-only always uses reconstruction error. P2-only always uses the combined reconstruction-error plus SVDD-distance score. Full uses reconstruction error in Phase 1 and the combined score in Phase 2. Run the ablation only on Fashion-MNIST with malicious-client ratios of 10%, 20%, 30%, and 40%, using LF, GN, SF, and BD with seeds 42--44. LF, GN, and SF report DAR / ACC; BD reports DAR / ASR.

Table 3. Two-stage ranking-schedule ablation on Fashion-MNIST.

Malicious ratio	Configuration	LF		GN		SF		BD	
		ACC	DAR	ACC	DAR	ACC	DAR	ASR	DAR
10%	P1-only	—	—	—	—	—	—	—	—
10%	P2-only	—	—	—	—	—	—	—	—
10%	Full	—	—	—	—	—	—	—	—
20%	P1-only	—	—	—	—	—	—	—	—
20%	P2-only	—	—	—	—	—	—	—	—
20%	Full	—	—	—	—	—	—	—	—
30%	P1-only	—	—	—	—	—	—	—	—
30%	P2-only	—	—	—	—	—	—	—	—
30%	Full	—	—	—	—	—	—	—	—
40%	P1-only	—	—	—	—	—	—	—	—
40%	P2-only	—	—	—	—	—	—	—	—
40%	Full	—	—	—	—	—	—	—	—

5.5 Computational Overhead

Measure overhead on the same hardware, software environment, client count, round count, and worker schedule. Exclude dataset download and repeat every measurement for all three seeds. Report median wall-clock time, sample standard deviation, peak allocated GPU memory, throughput, and result-file size where available.

For AE-SVDD, separate client local training, descriptor construction, AE/SVDD update, validation candidate evaluation, and aggregation. Evaluation of all internal candidate ratios is part of the method cost and must not be omitted.

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TABLE V
COMPUTATIONAL OVERHEAD OF DIFFERENT DEFENSE METHODS

Detector	MNIST	EMNIST	CIFAR10
Multi_krum	0.73s	0.19s	75.31s
Auror	1.45s	0.56s	128.28s
FoolsGold	0.87s	0.21s	59.91s
FLDetector	6.55s	6.31s	160.51s
FedDMC	0.98s	0.34s	64.72s

The best results are bold.

All conclusions must be based on complete result files with the expected round count. Failures are summarized separately in the reproducibility appendix rather than silently removed.